

For questions 1-2, consider the following context.

After several power outages caused by storms, Phyllis purchased a used generator for \$1,995 four years ago. Equipment loses value each year from the basic wear and tear and as newer models come on the market. Phyllis estimates the generator loses about 15% of its value each year based on the prices she sees online.

- Phyllis uses the expression $1995 \cdot 0.85^4$ to determine a fair price for which to sell the generator now. What is the fair price Phyllis determined?

$$1,995 \cdot 0.52200625 = \$1,041.40$$

- If Phyllis' estimation of how much value the generator loses each year is accurate, she can use the expression $1995 \cdot 0.85^{-2}$ to determine the price of the generator two years before she bought it when it was new. Evaluate the expressions to find the price of the generator when it was new.

$$1,995 \left(\frac{1}{0.7225} \right) = \$2,761.25$$

For questions 3-4, consider the following context.

Janet is considering planting cabbage for the first time on her farm. She has an acre set aside and wants to determine whether the costs and potential profits make sense. After researching, she determines the following information.

Costs of Planting 1 Acre of Cabbages	Expected Revenue from 1 Acre of Cabbages
- Plants: \$1800 - Labor: \$675 - Materials: \$1295	- Total Plants: 12,000 - Expected percentage available for sale: 75% - Average revenue per plant: \$3.58

The expression $3.58(0.75 \cdot 12,000) - (1800 + 675 + 1295)$ can be used to find the amount of money Janet can expect to earn based on her research. Profit can be calculated using the equation Profit = Revenue – Costs.

- Which part of the expression that Janet wrote represents the cost of planting 1 acre of cabbages? $(1,800 + 675 + 1295)$
- Which part of the expression that Janet wrote represents the revenue of planting 1 acre of cabbages? $3.58(0.75 \cdot 12,000)$
- Determine the profit Janet is expecting to earn based on her research.

$$3.58(9,000) - 3,770 \rightarrow 32,220 - 3,770 = \$28.450$$

For questions 6-10, consider the following context.

Dominic is looking to do some home renovations on his condo. He plans to re-tile the floor of the bathroom and lay new carpeting on the floor of the bedroom, including the closet. Dominic's bedroom, including the closet, measures 12 feet (ft) by 12 ft, and his bathroom measures 7 ft by 10 ft.

The prices Dominic found for the materials he needs to update the flooring in both rooms is given in the table.

	Bedroom		Bathroom		
	Carpet	Padding Material	Underlay	Tile	Grout
	Home Decorators Sycamore	Memory Foam	OSA	Marazzi	Polyblend
Price	\$3.20/ft ²	\$0.87/ft ²	\$1.55/ft ²	\$1.95/ft ²	\$13.56/ft ²

Dominic writes the expression $12^2(3.20 + 0.87 + 1.55) + (7 \cdot 10)(1.95) + 13.56$ to determine the total cost of flooring both rooms.

6. Which part of the expression represents the cost of carpeting the bedroom?

$$12^2(3.20 + 0.87 + 1.55)$$

7. Which part of the expression represents the cost of tiling the bathroom?

$$(7 \cdot 10)(1.95) + 13.56$$

8. Evaluate the expression to determine how much it will cost Dominic for the materials to replace the flooring in both rooms.

$$144(5.62) + (70)(1.95) + 13.56 \rightarrow 809.28 + 136.5 + 13.56 = \$959.34$$

9. Dominic is concerned about the total cost and wants to know how much it would cost to re-floor the bedroom only. How much would the flooring materials cost for only his bedroom?

$$12^2(3.20 + 0.87 + 1.55) \rightarrow 144(5.62) = \$809.28$$

10. Dominic is concerned about the total cost and wants to know how much it would cost to re-floor the bathroom only. How much would the flooring materials cost for only his bathroom?

$$(70)(1.95) + 13.56 \rightarrow 136.5 + 13.56 = \$150.06$$